

ENGR 095

Fall 2026 - Jenn Winikus & Nick Strandwitz

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Sensing the World - Materials Engineering Reflection 1

In this lab module, we learned what specific materials engineers and computer engineers do and focused on building a sensor. To do this, we first learned what a sensor was, which is something that takes in input to produce an output. In the Wilbur Powerhouse Lab, we were able to use a vinyl cutter to cut out a thin sheet of yellow vinyl on backing tape. This vinyl had a specific pattern of an interdigitated electrode (IDE). We then took this IDE into Whitaker and made more electrode devices on glass slides using tape and cutting out patterns ourselves with an x-acto knife. We learned what resistivity was, which is how strong a material can resist electricity, and what resistance is, which is how much a specific object can resist electricity based on its size. We measured a piece of copper tape resistance and then went back to our sensors. As groups of 3, we each picked a metallic material which ranged from silver (Ag) paste, silver (Ag) paint, and carbon paint. Taking our specific material and the IDEs we made before, we deposited a large amount of the metallic material onto the glass slide, and used another slide to scrape off the excess material while filling in the parts of the vinyl or tape that was cut out. We measured the resistivity in each material as a base line. After, we used Poly ethylene dioxythiophene, poly styrene sulfonate (PEDOT-PSS) which is a conducting/semiconducting polymer and spin coated our sensor IDEs. By spin coating, it creates a thin film over our IDEs which completes our sensors by allowing it to be conductive. We then could test the resistivity of our sensors after breathing on it to create humidity and seeing how much the resistivity would change.

Through this experience, I learned a lot about material science. Materials science was something I had never really known much of. But with this, I learned a lot of how much engineering relies on materials engineers to know the right kind of materials to use. I was able to learn a lot more about resistivity and resistance and did a lot of calculations with it. I however, found myself lost a lot and unsure of what to do. It seemed like everyone already knew materials that were conductive or not, how to calculate resistance and resistivity, and knew what was going on. I felt a lot of the time but with the help of my partners and asking questions to the TAs and instructors, I found myself slowly gaining confidence and understanding in what we were doing. I enjoyed when we deposited material onto the IDEs and scraped it off, as well as the spin coating part. I did not enjoy the calculations much, though some of it wasn't that bad. Overall, this part of the module was something I would have never done before or experienced and I'm glad to have experienced.